Formation of Wind

Air moves from high pressure to low pressure — that flow is wind, and fast winds lower the pressure further.

Why air moves

Air moves from a region of high pressure to a region of low pressure. This moving air is **wind**. The bigger the pressure difference, the faster the wind.

Sea breeze and land breeze

	Day — Sea breeze	Night — Land breeze
Heats faster	Land	(Land cools faster)
Low pressure over	Land (warm air rises)	Sea (warmer water)
Wind blows from	Sea → Land	Land → Sea

High-speed winds lower the pressure

Where air moves fast, its **pressure drops**. Blow between two hanging balloons and they move **towards** each other — the fast air between them is low pressure, and higher pressure outside pushes them in.

> **Why roofs blow off:** high-speed wind over a roof creates low pressure above it; the higher pressure below can lift a weak roof. Keeping doors/windows **open** during a storm reduces this pressure difference and protects the roof.

Where students lose marks

> **Trap 1 — Wind blows from HIGH to LOW pressure.** Not the other way round.

> **Trap 2 — Fast wind = LOW pressure.** High-speed air has reduced pressure, which is why things get pushed towards it.

> **Trap 3 — Open windows in a storm.** Open (not shut) windows equalise pressure and help keep the roof on.

> **Breezes:** sea breeze blows sea→land in the day; land breeze blows land→sea at night.

Model answers — write them like this

Example A. Why does a sea breeze blow during the day?

Step 1: Land heats faster than the sea; warm air over land rises, making low pressure there.

Step 2: Air moves from high pressure (sea) to low pressure (land).

∴ Wind blows from sea to land — a sea breeze.

Example B. Two balloons hang with a gap; you blow between them. Why do they come together?

Step 1: Fast-moving air between them creates a low-pressure region.

Step 2: Higher pressure outside pushes the balloons inward.

∴ The pressure difference pushes them towards each other.

Ready to practise? → Open the worksheet